

SOLAR CYCLE ANALYSIS

Sunspot number Time-series



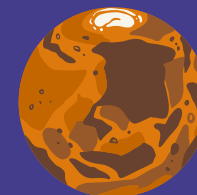
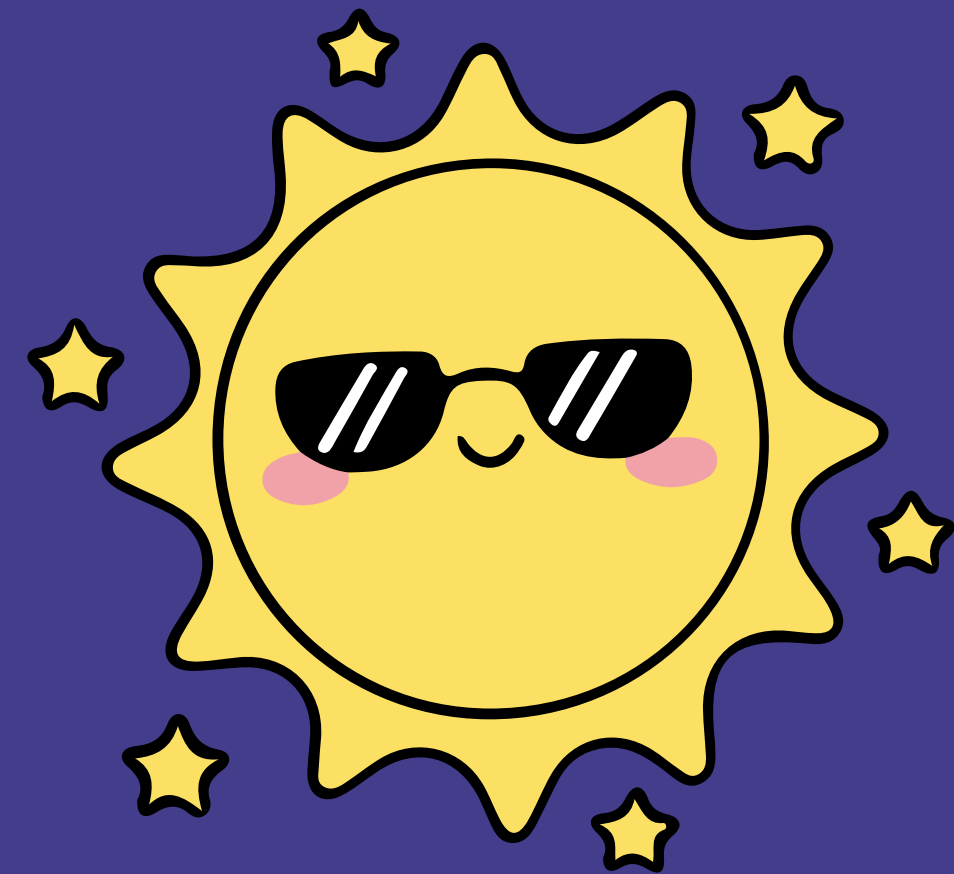
Presented by

Group 2

Isabella DeAbreu, Laurenz Jakob Kluth, Navenkumar Balasubramaniam, Sofia Mollon Laorca, and Vikram Bhatt

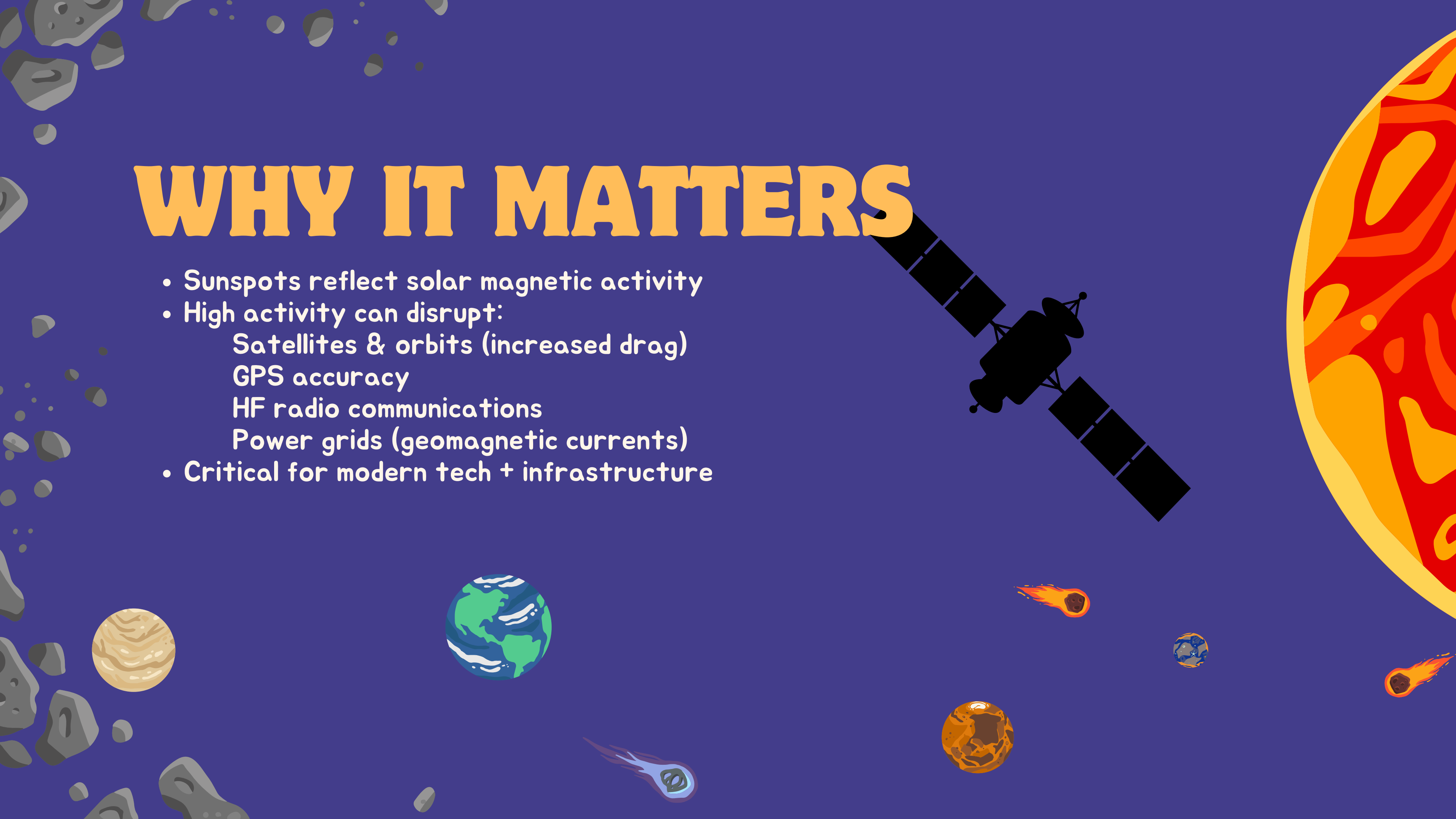
WHAT DATA ARE WE SEEING?

- Monthly sunspot numbers from NOAA SWPC / S.I.D.C.
- Long record: 1749–2025
- Fields used: time-tag (YYYY-MM), ssn
- Shows the repeating ~11-year solar cycle



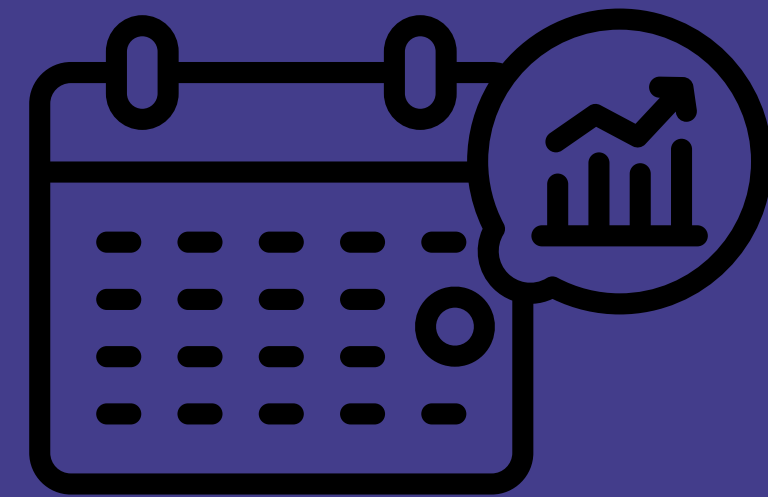
WHY IT MATTERS

- Sunspots reflect solar magnetic activity
- High activity can disrupt:
 - Satellites & orbits (increased drag)
 - GPS accuracy
 - HF radio communications
 - Power grids (geomagnetic currents)
- Critical for modern tech + infrastructure



HOW PREDICTION HELPS

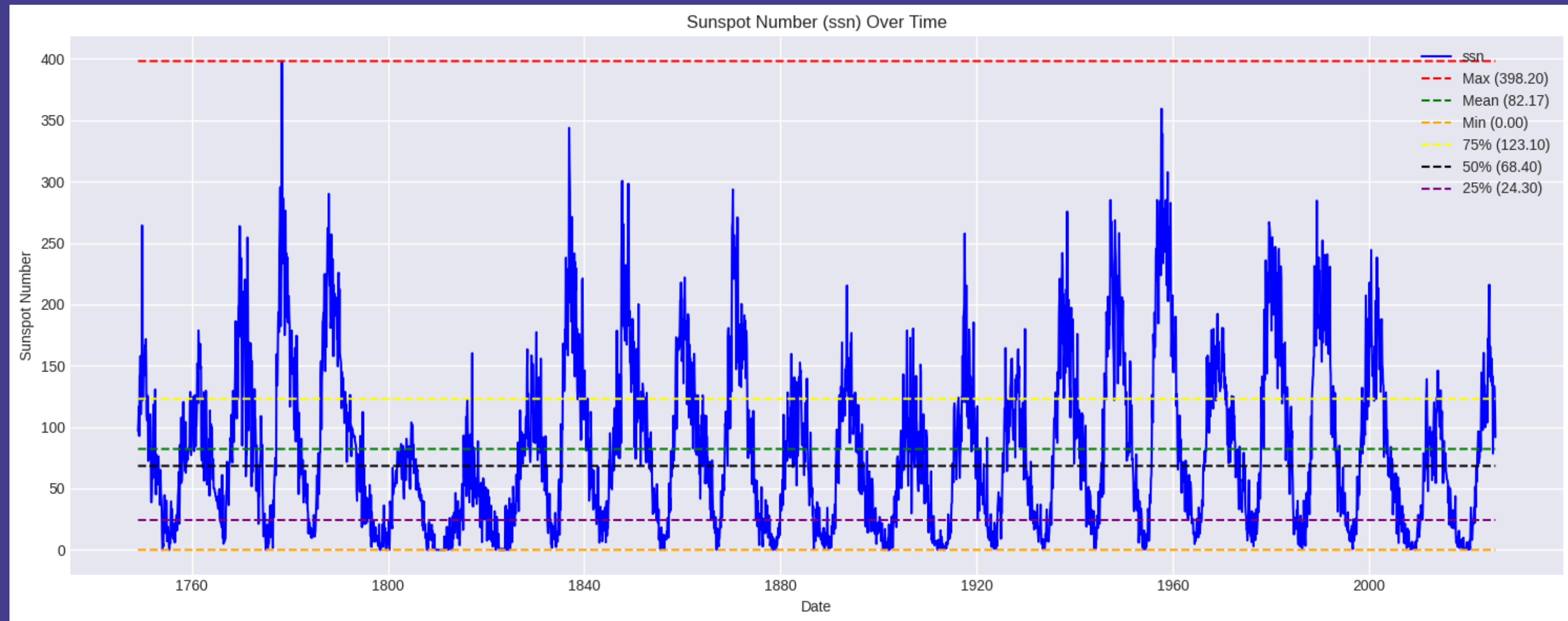
- Forecasts give advance warning of active periods
- Enables preventive measures: satellite adjustments, grid protection, mission planning
- Helps reduce operational risk and improve resilience
- Supports better decision-making during high solar activity



EXPLORATORY ANALYSIS

Main Patterns & Trends

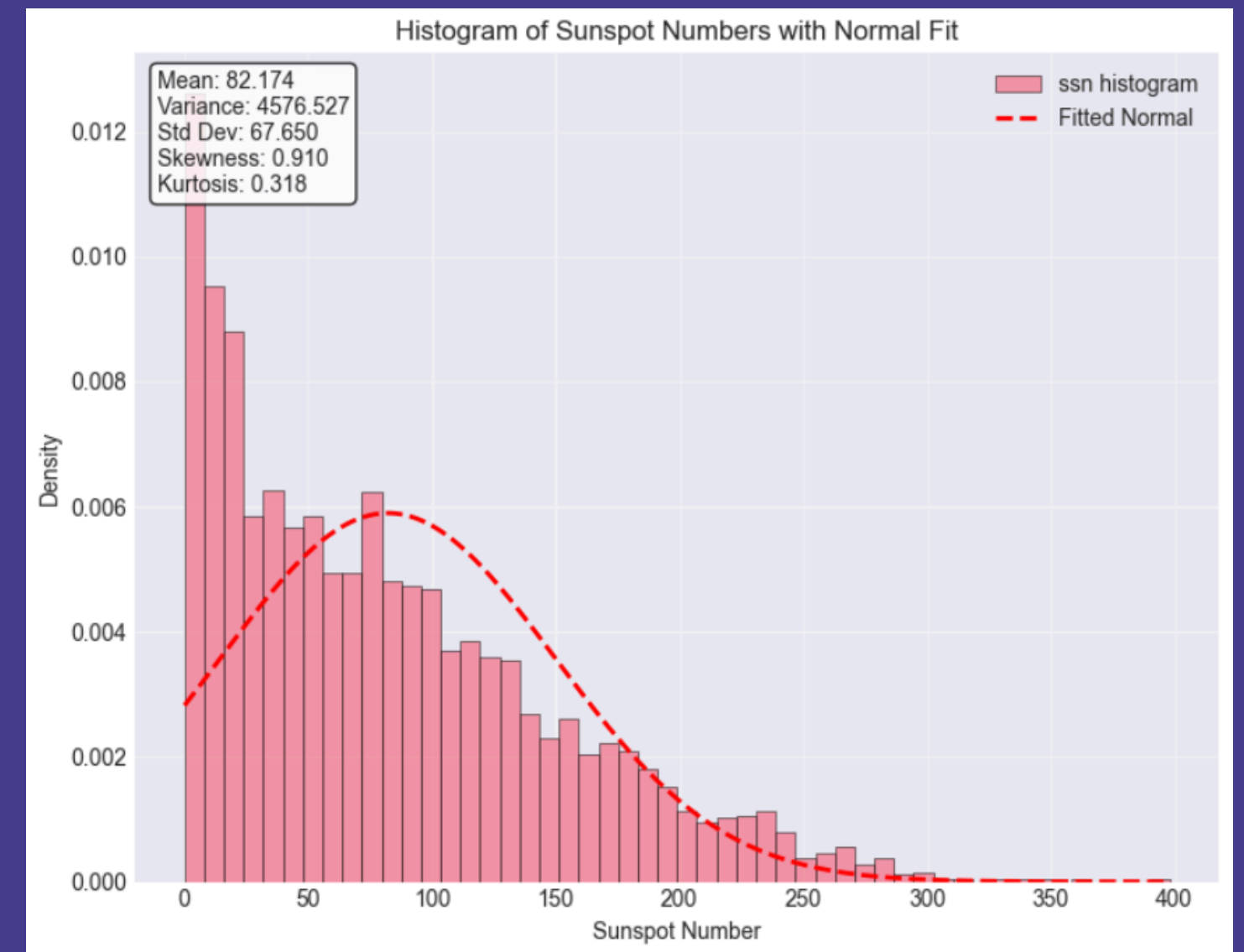
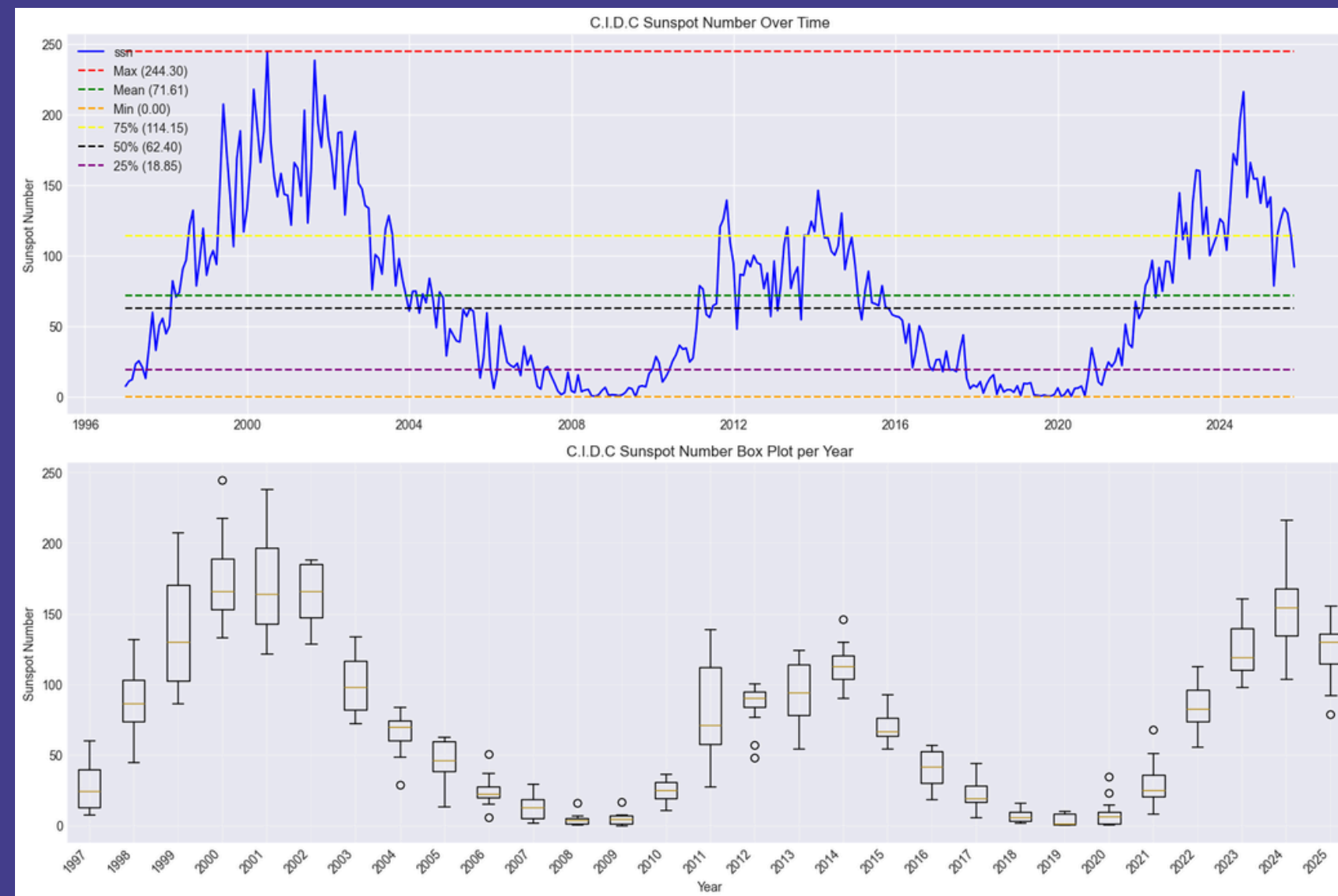
Confirmation: Clear repeating 11-year solar cycle with no long-term trend, but strong seasonal structure



EXPLORATORY ANALYSIS

Main Patterns & Trends

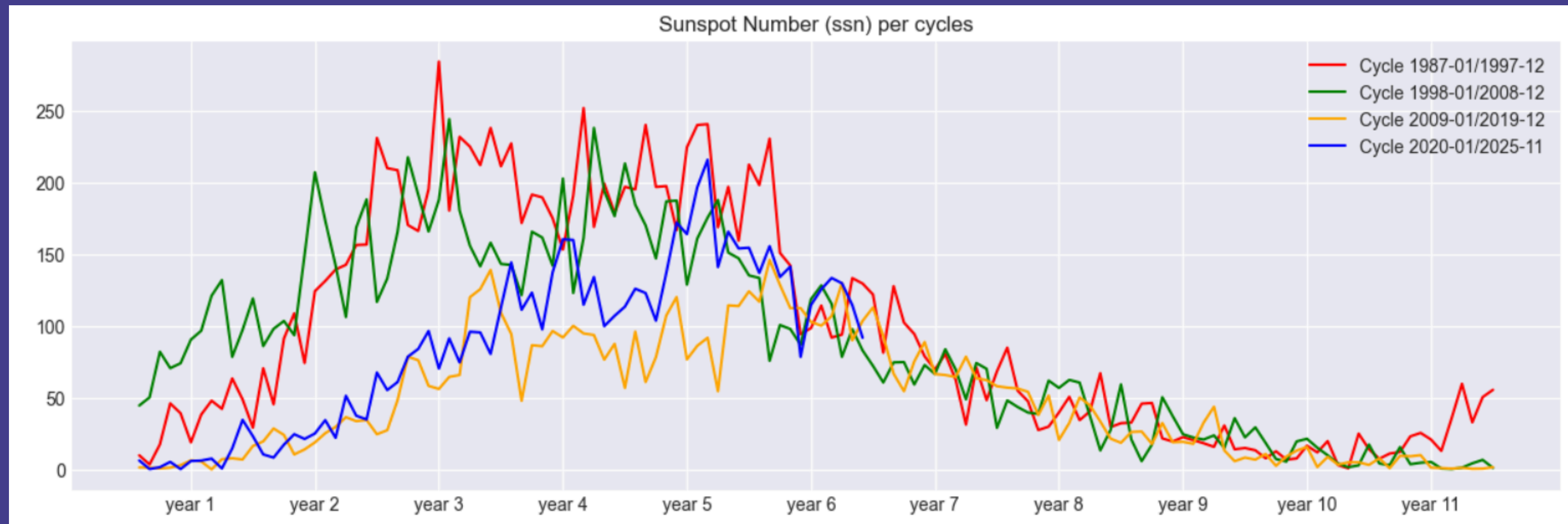
Observation: Right-skewed & heavy-tailed distribution for total data



EXPLORATORY ANALYSIS

Main Patterns & Trends

Pattern: Similarities in mean and variance when comparing cycles, question of stationarity as cycles are not irregular and tied to time

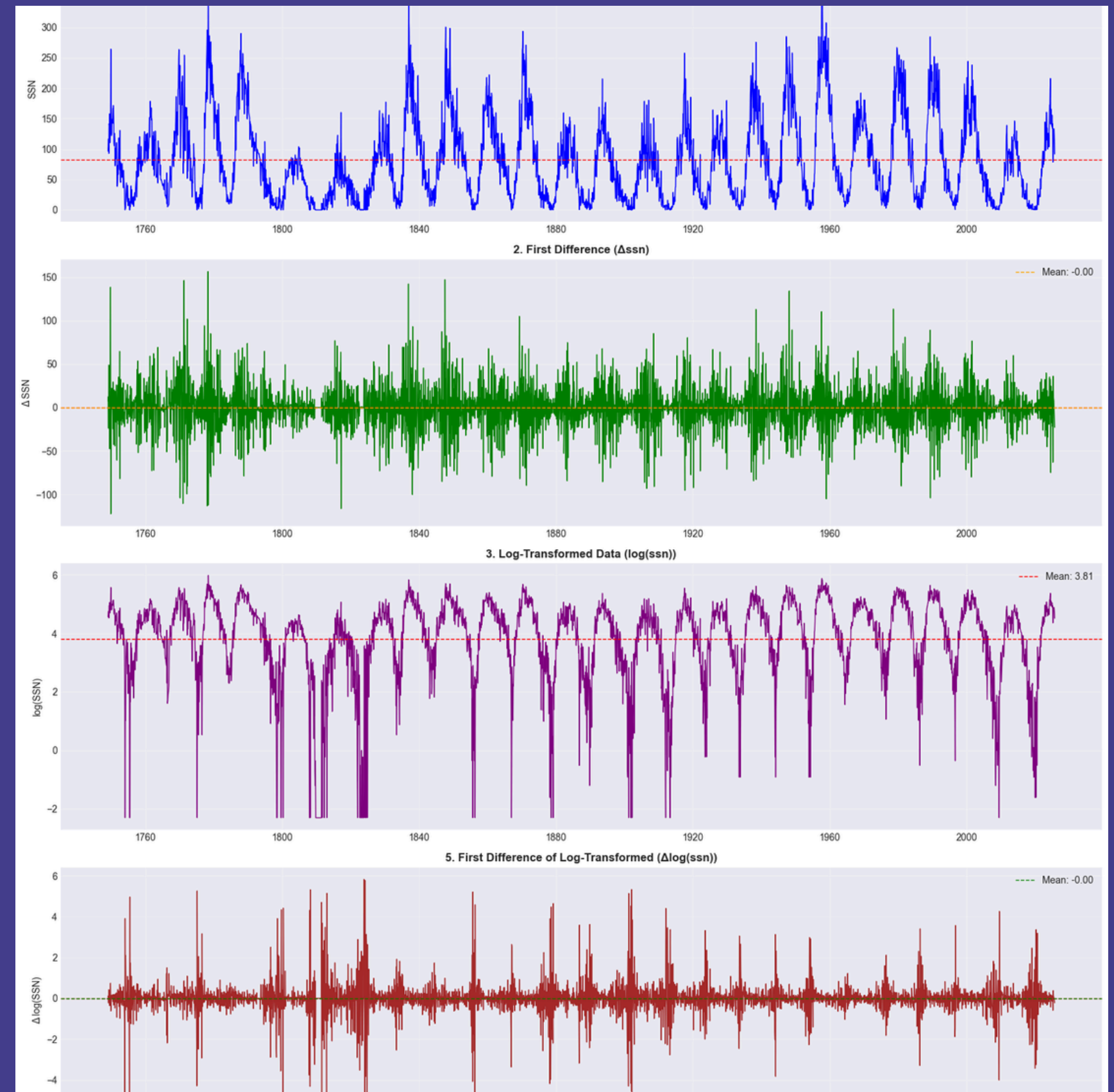


EXPLORATORY ANALYSIS

Differencing and Stationarity

- KPSS and ADF test show stationarity for all data types.
- Keep to original and first differenced data

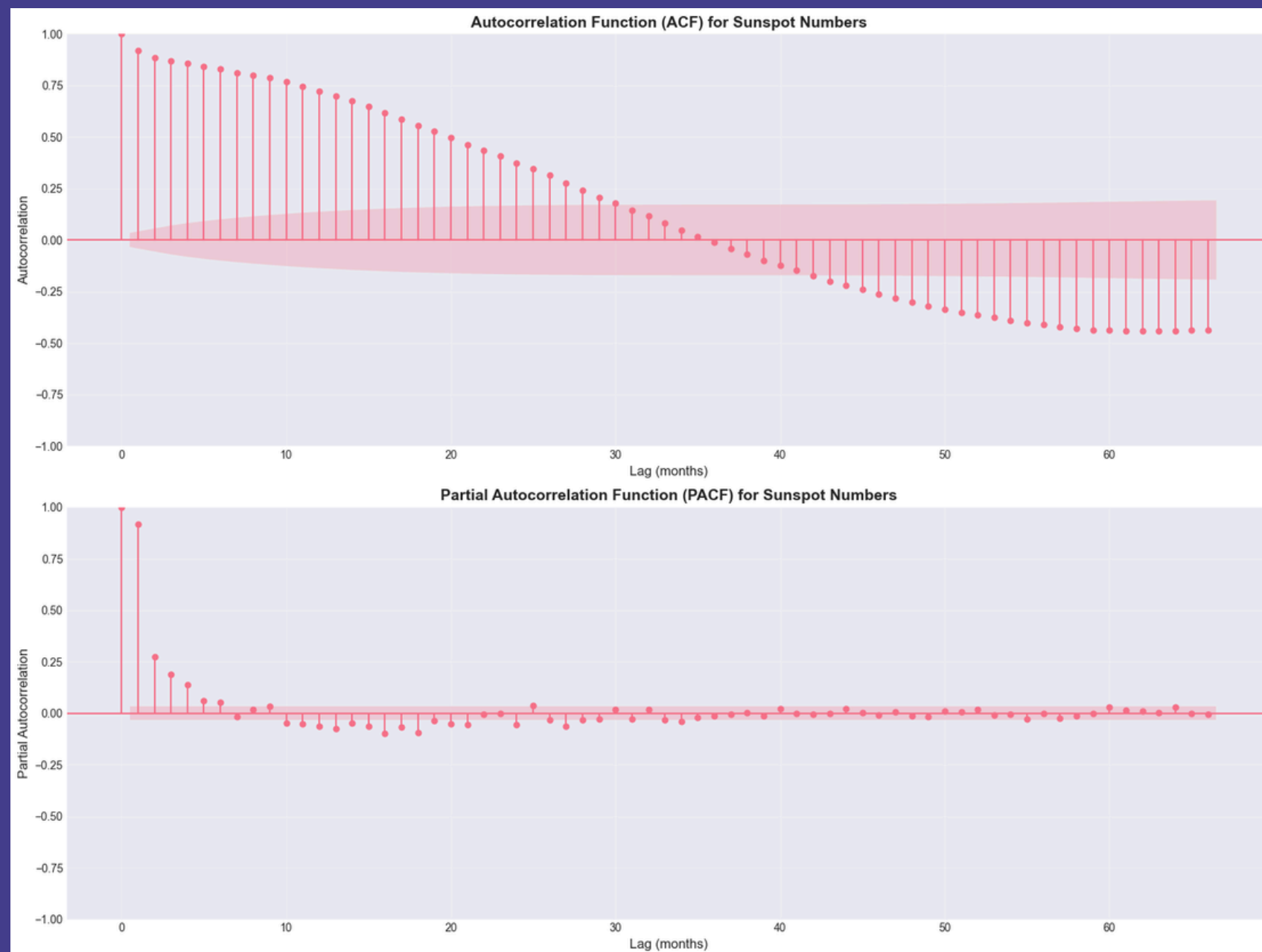
	KPSS Test	ADF Test
Original Data (ssn)	0.1	• 3.54×10^{-19}
FIRST DIFFERENCE (Δ ssn)	0.1	• 2.51×10^{-15}
LOG-TRANSFORMED DATA (log(ssn))	0.1	• 5.38×10^{-15}
FIRST DIFFERENCE OF LOG-TRANSFORMED DATA (Δ log(ssn))	0.1	• 4.72×10^{-23}



EXPLORATORY ANALYSIS

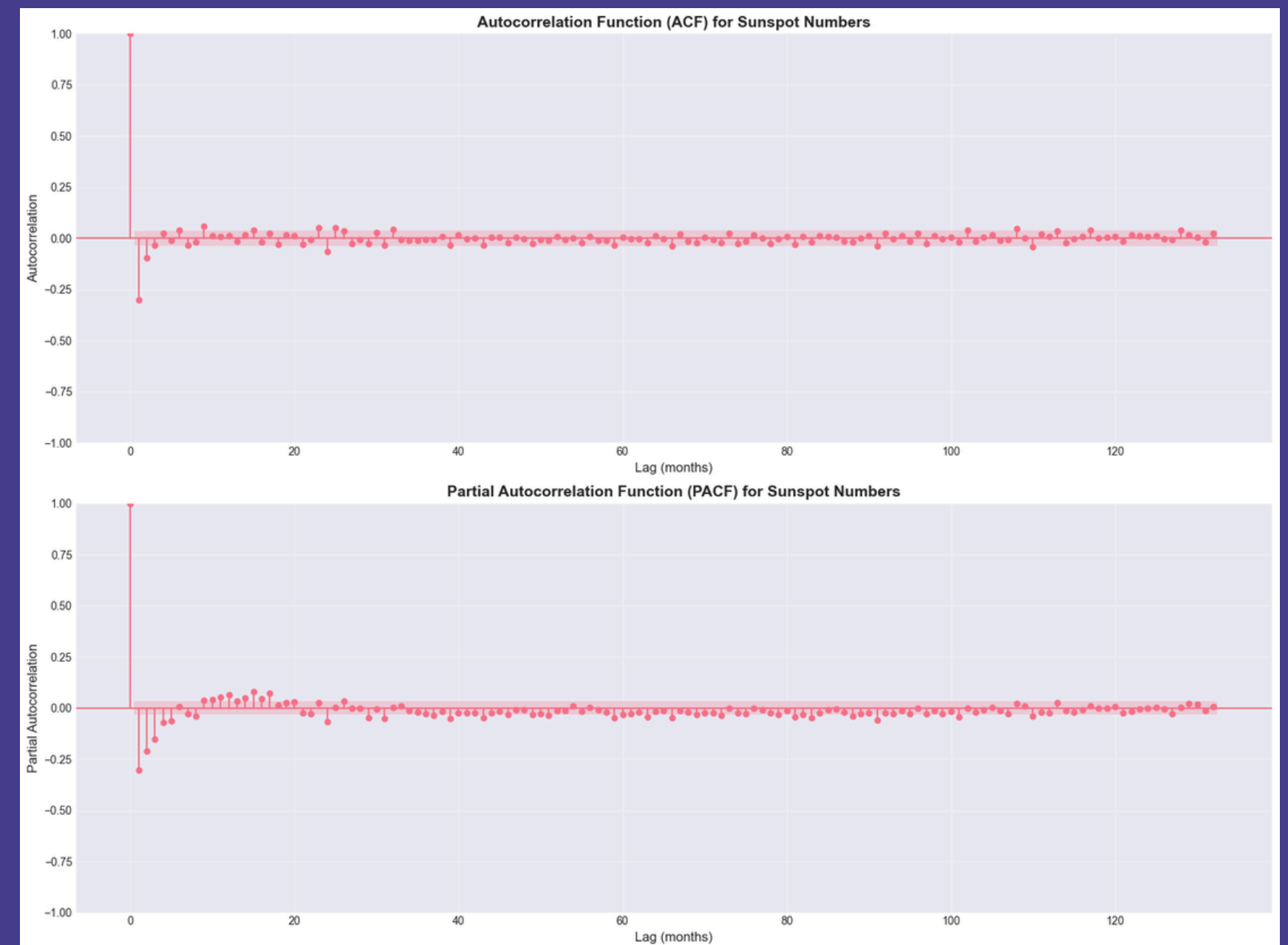
Regular Data

- ACF → Slow decay + oscillation
- PACF → Strong early lags, AR structure.



Differentiated data

- ACF → Sharp cut off at lags 1 and 2 AR
- PACF → Strong early lags in 1 and 2 MA. Successful in removing nonstationary.



MODEL EVALUATION

Non-seasonal ARIMA (2,0,2):

- Reasonable baseline
- Still showing seasonal autocorrelation (undermodeling)

Non-seasonal ARIMA (1,0,1)

- Higher AIC/BIC and MAE/MAPE values (doesn't capture long-term or short-term structure)

Non-seasonal ARIMA (2,1,2):

- Best fit for nonseasonal ARIMA's with the lowest error metrics
- Differencing added but leaves autocorrelation

Non-seasonal ARIMA (1,1,1)

- Higher AIC/BIC than ARIMA (2,1,2) and weaker error metrics
- Residuals have remaining autocorrelation

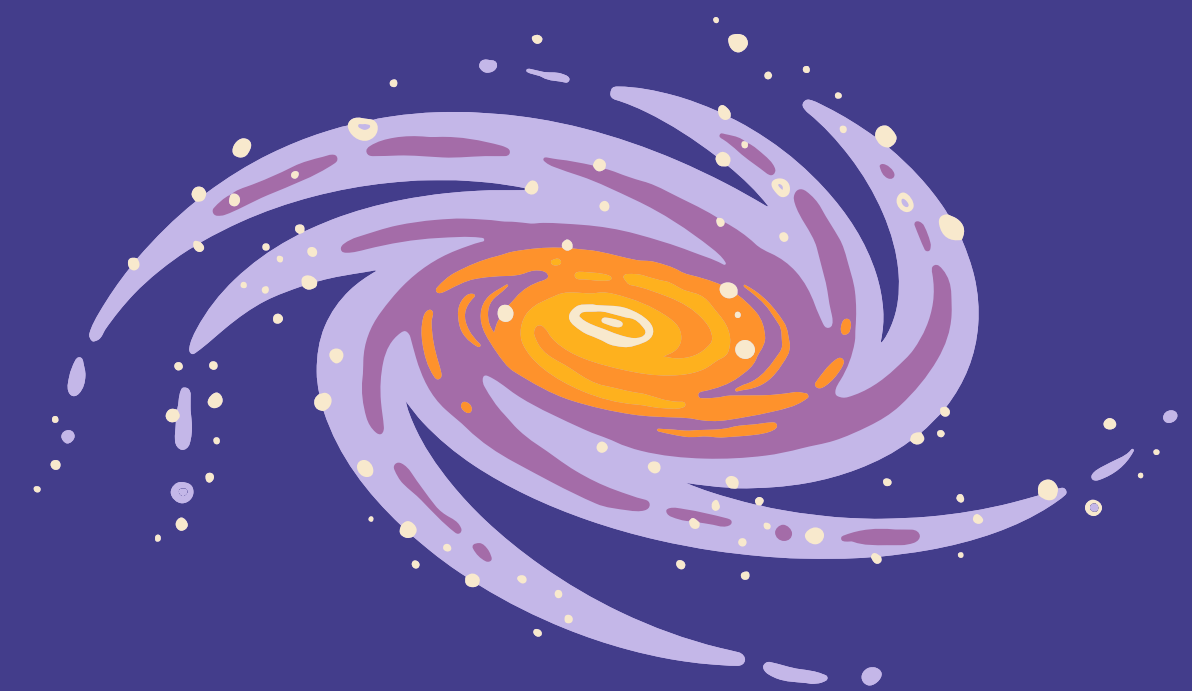
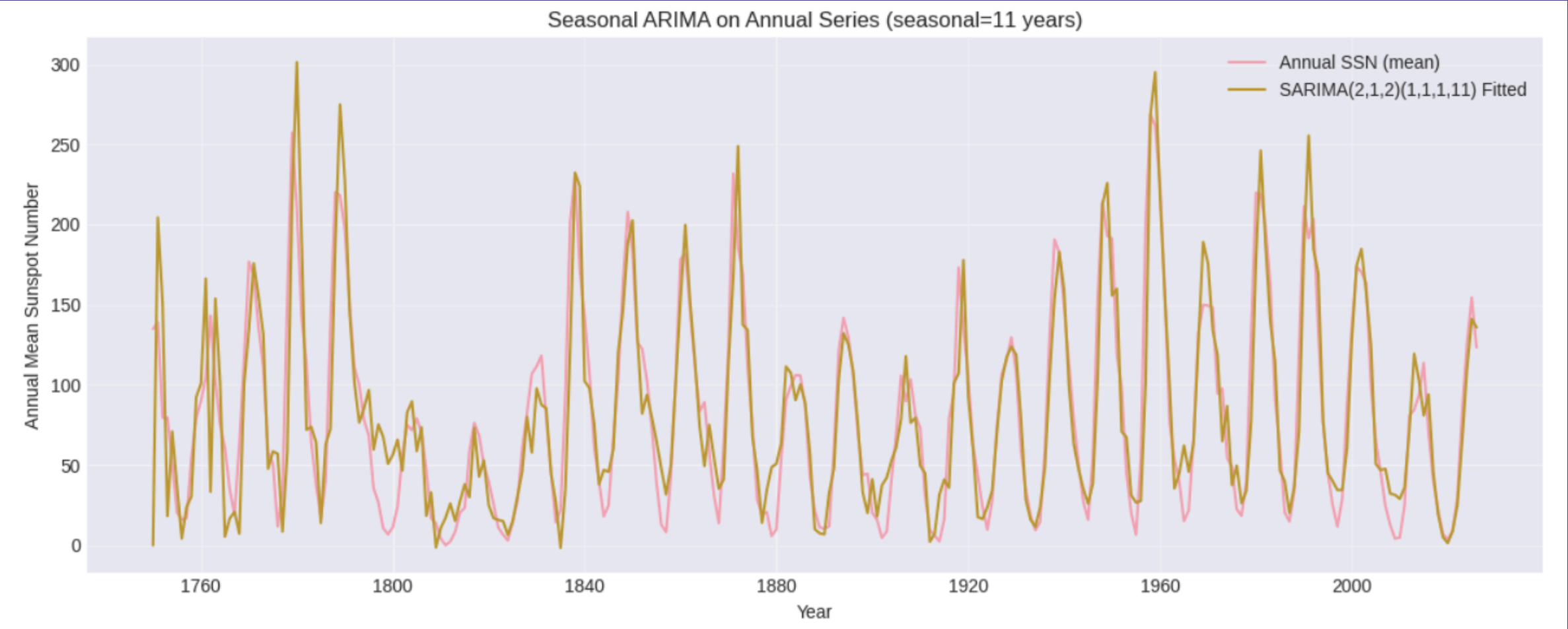
SARIMA with 11 years (132-month) seasonality:

- Best AIC/BIC of all models (lowest)
- Residuals $\approx$ white noise (no remaining autocorrelation)
- Similar error metrics to ARIMA models, although not the lowest
- Captures true physical solar cycle
- Best for forecasting (long-term)

Takeaway:

- Model accuracy substantially increases when acknowledging true solar periodicity of 11 years (132 months), not annual patterns.

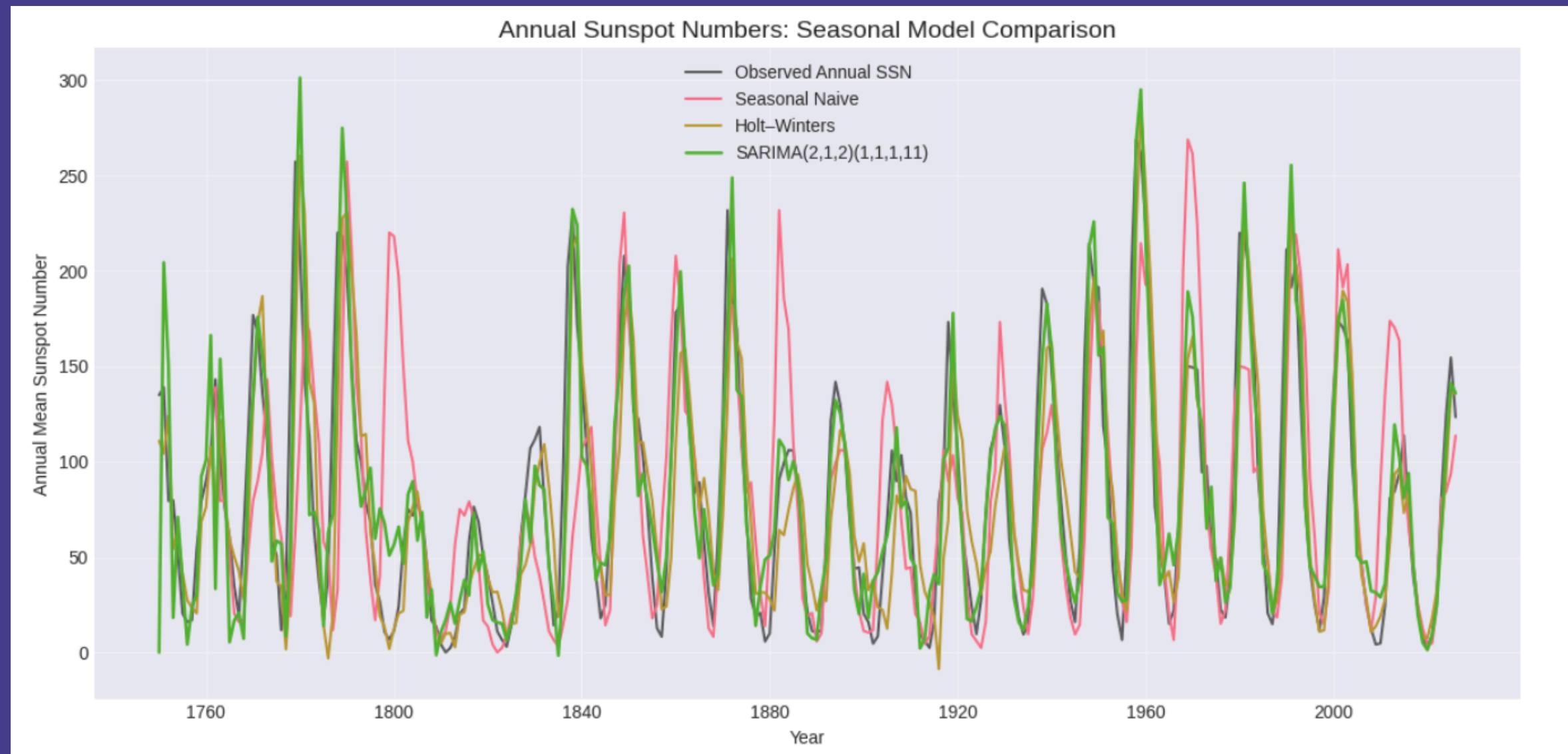
MODEL SELECTION AND ERROR METRICS



11-year Seasonal ARIMA error metrics show that incorporating the 11 year solar cycle period greatly improves model structure and long-term accuracy, outperforming nonseasonal ARIMA models despite slightly higher short-term errors.

	AIC	BIC	RMSE	MAE	MAPE	MASE
Model						
ARIMA(2,0,2)	2604.703	2626.447	25.029	20.408	73.082	0.692
ARIMA(1,0,1)	2691.338	2705.834	25.202	23.615	74.485	0.801
ARIMA(2,1,2)	2575.133	2593.235	24.882	19.561	56.673	0.664
ARIMA(1,1,1)	2702.348	2713.210	25.204	24.218	60.819	0.822
SARIMA(2,1,2)(1,1,1,11)	2365.651	2390.329	29.475	21.567	65.362	0.732

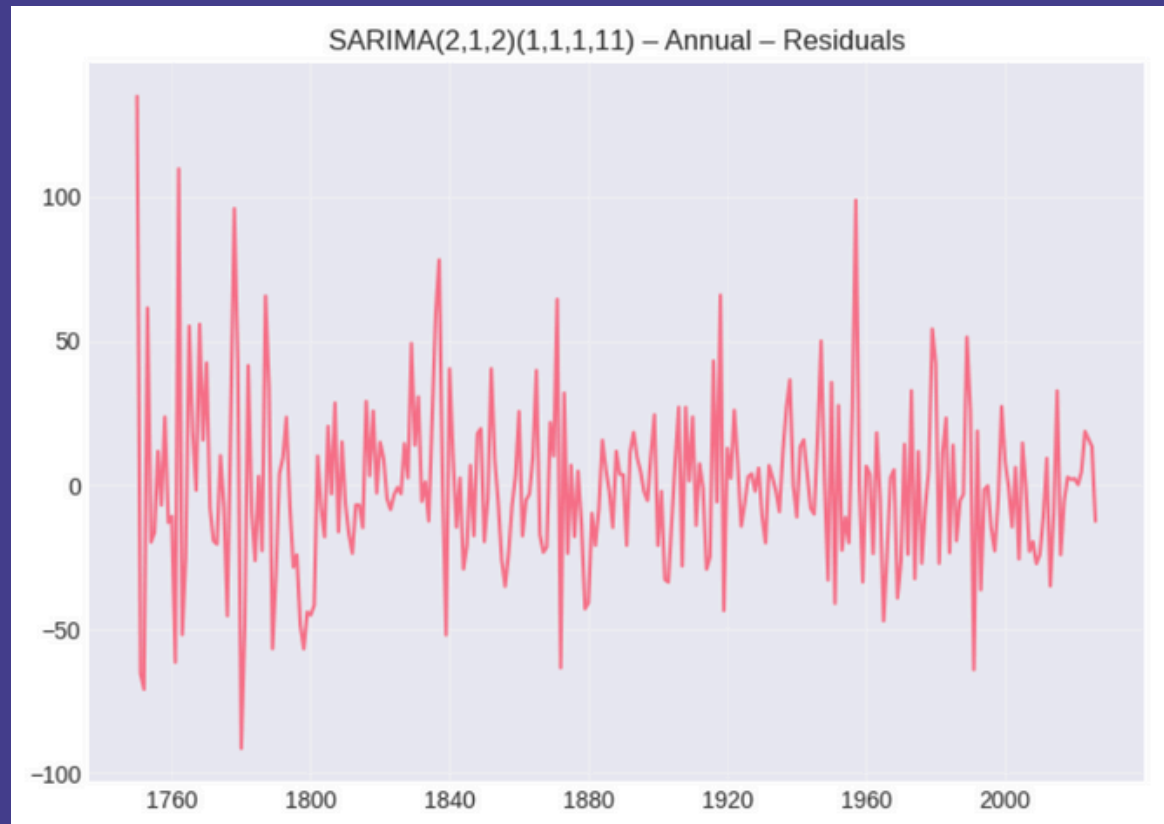
MODEL SHOWCASING



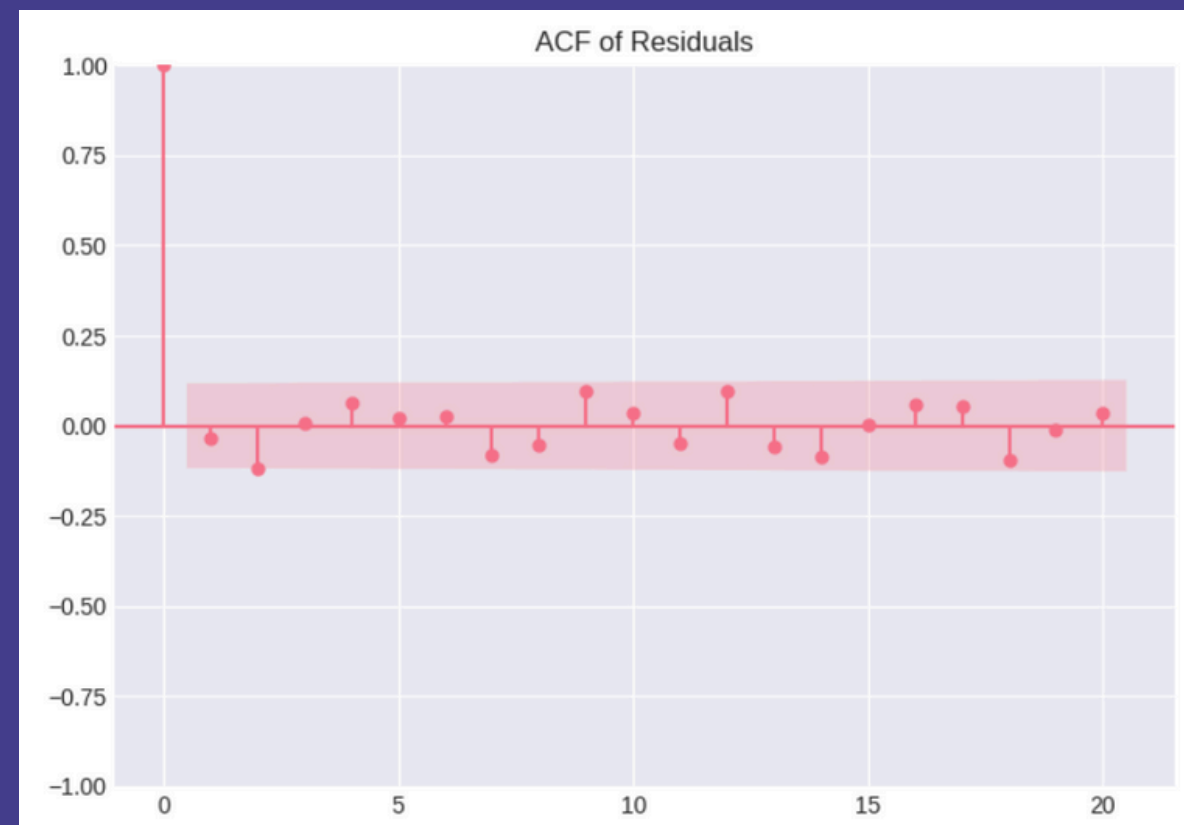
- SARIMA (11 year) provides the closest fit to the true solar cycle, outperforming the Seasonal Naive and Holt-Winters models in capturing annual sunspot patterns. Among the three models, SARIMA more closely replicates the observed annual sunspot curve, especially during cycle peaks.

RESIDUALS

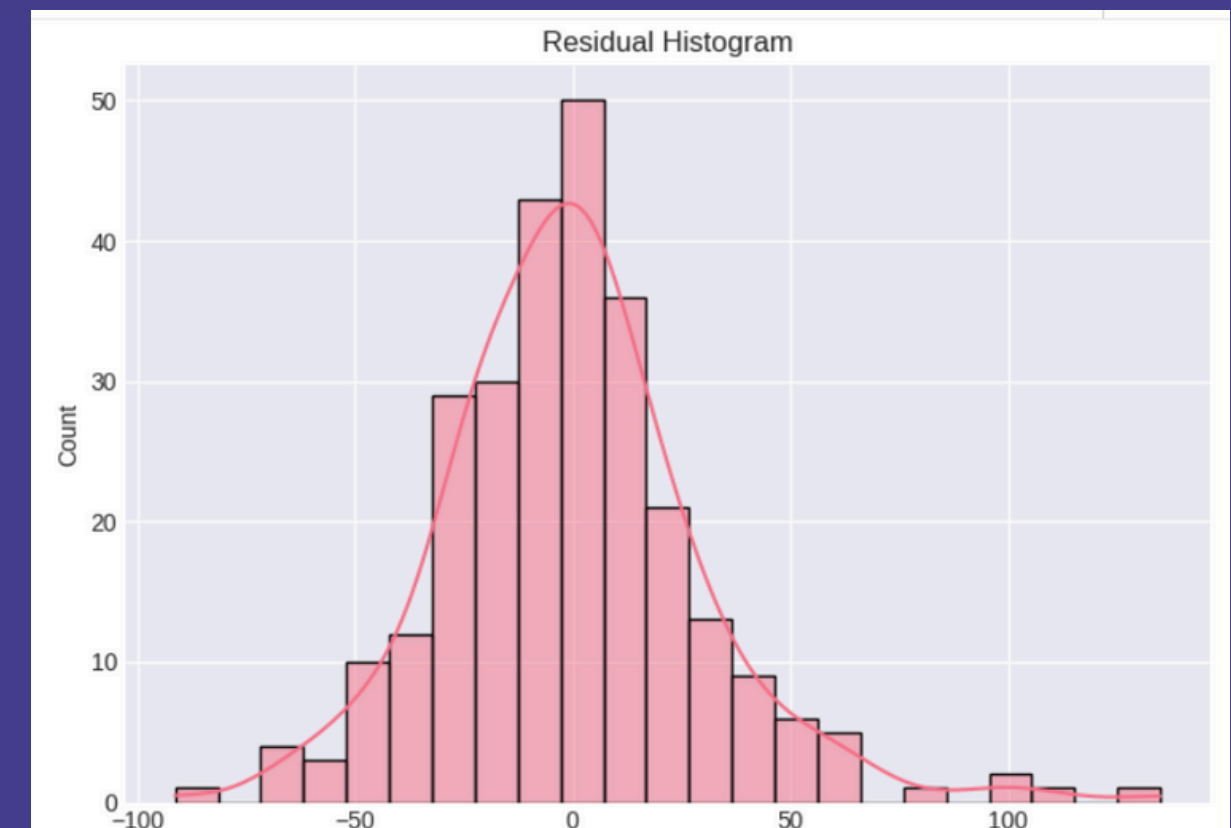
The 11-year Seasonal ARIMA residuals behave like white noise, showing a stable variance, no remaining autocorrelation, and an approximately normal distribution:



Mean & Variance



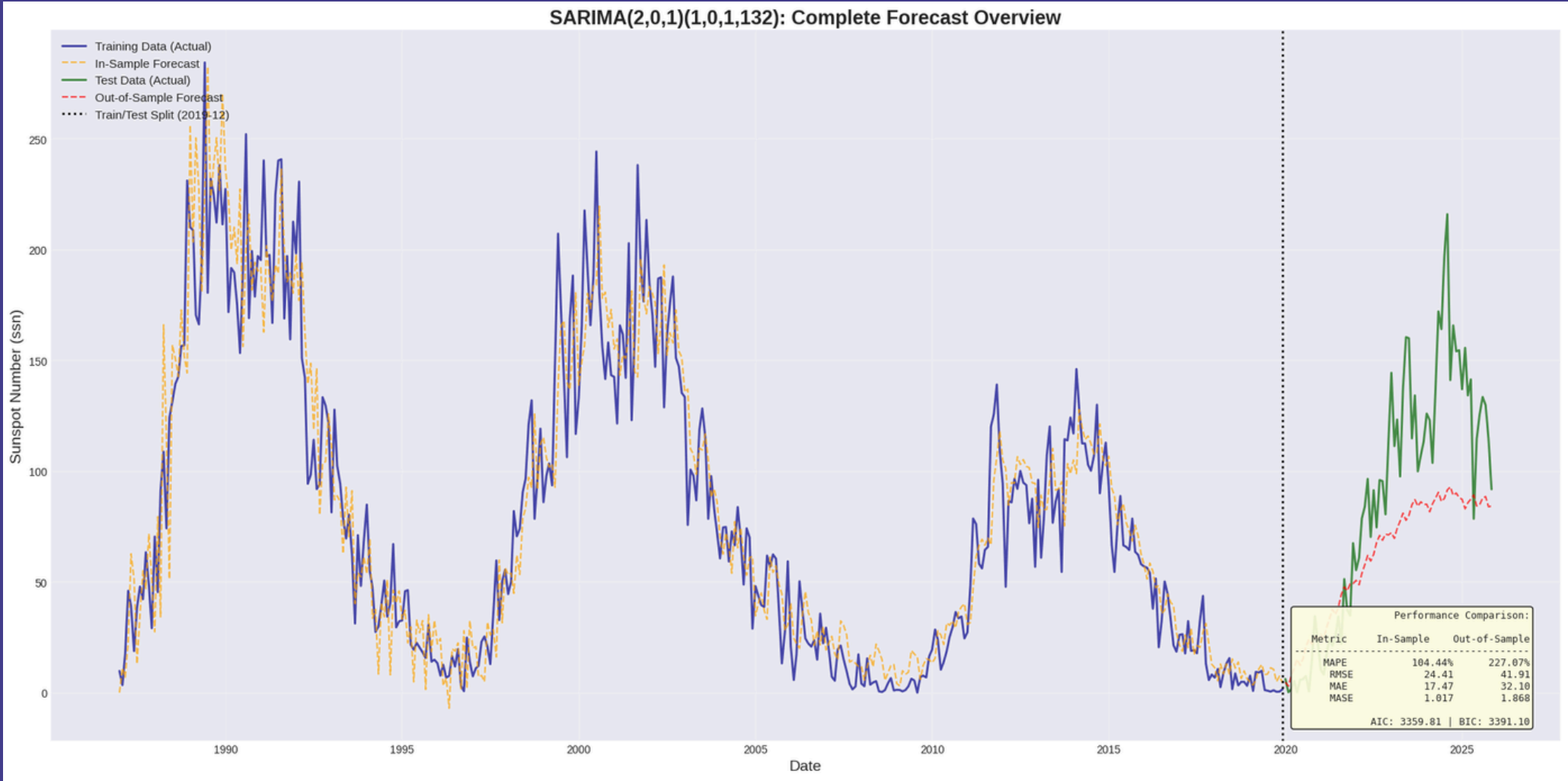
Uncorrelated



Normally distributed

TRAIN/TEST SPLIT

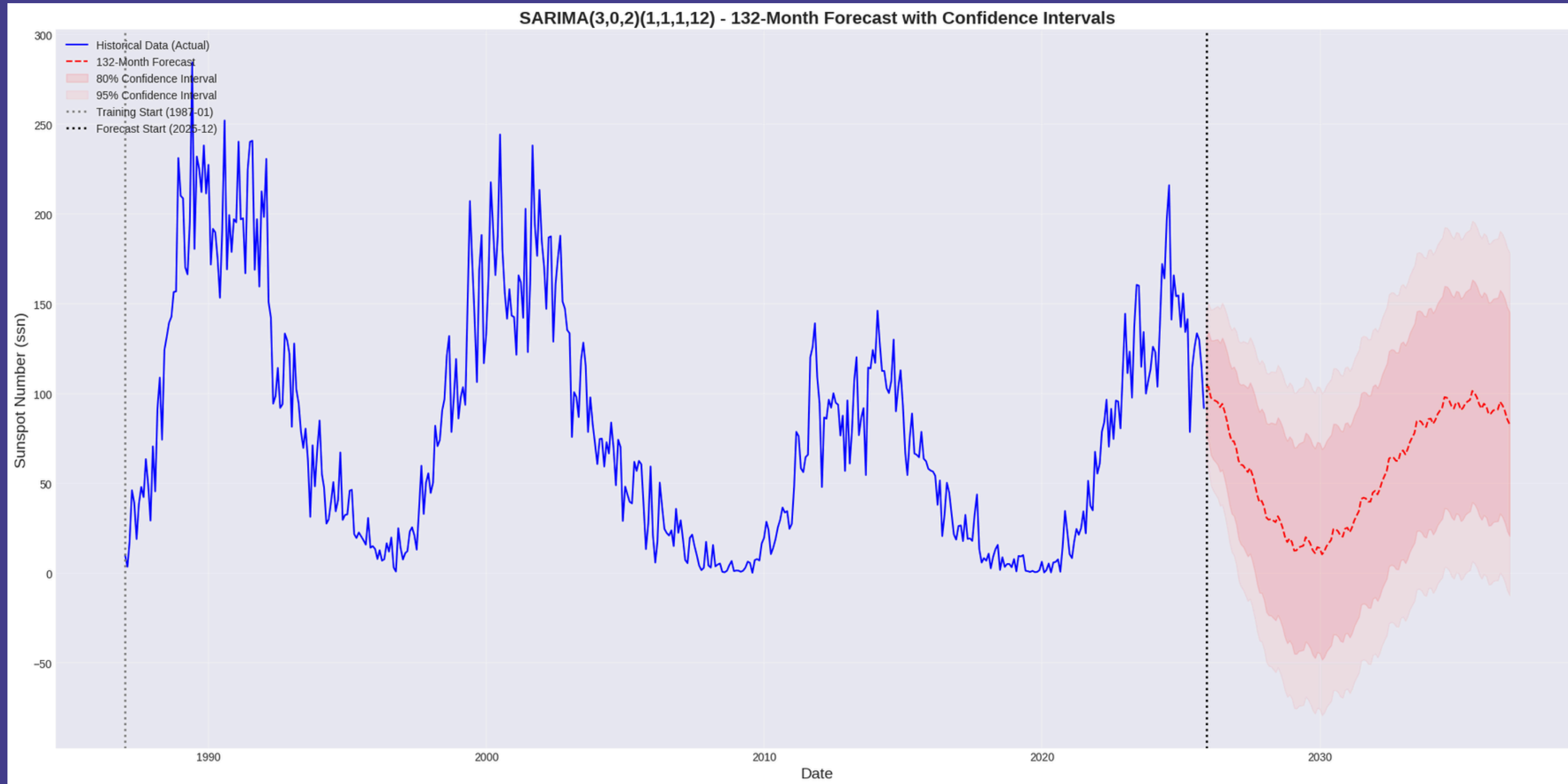
Forecast accuracy metrics for train and test split...



Performance Comparison:		
Metric	In-Sample	Out-of-Sample
MAPE	104.44%	227.07%
RMSE	24.41	41.91
MAE	17.47	32.10
MASE	1.017	1.868
AIC: 3359.81 BIC: 3391.10		

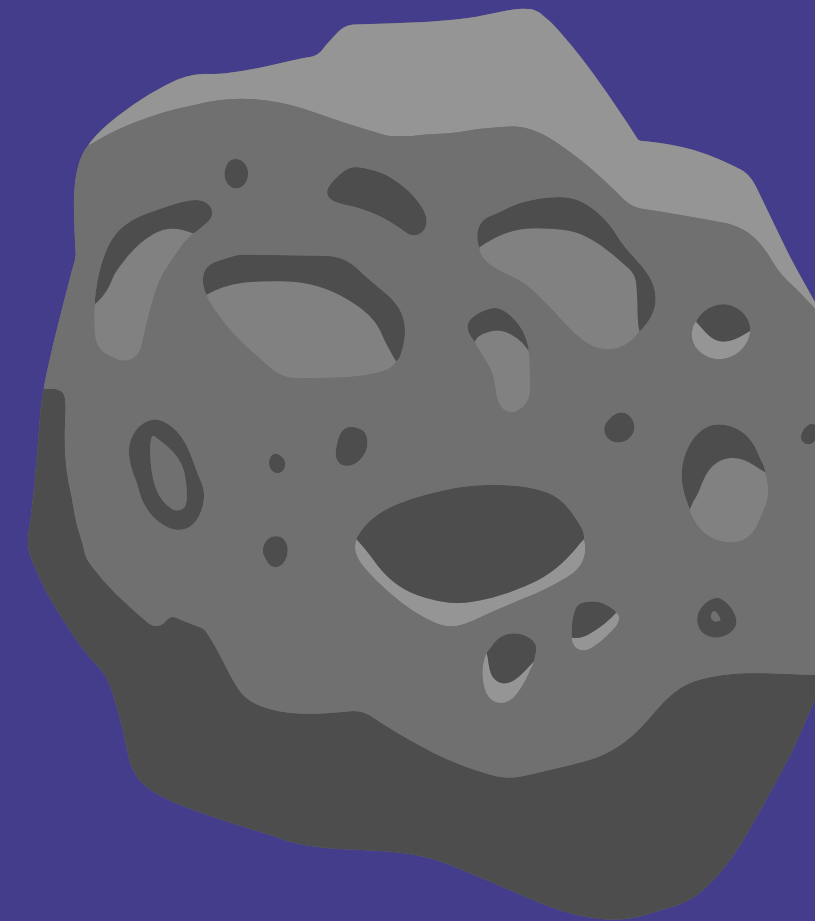
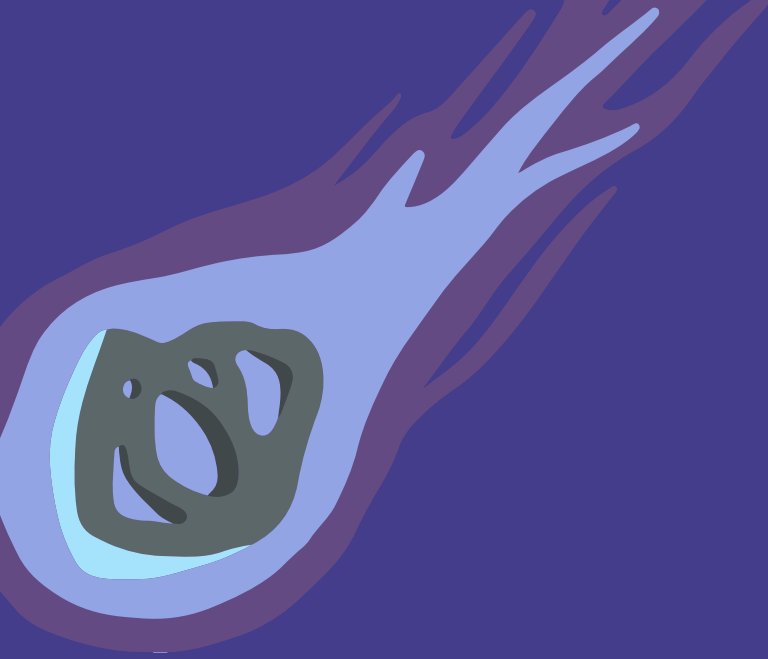
FORECASTING

In sample



KEY INSIGHTS

- Easy to predict cycle, hard to predict ssr over time.
- In sample prediction match well with cycles, out sample become harder to predict and lead to lower accuracy metrics.
- Cycles are uneven. Some are extremely high, while some stay lower.





**THANK YOU
FOR YOUR
ATTENTION**



**DO YOU HAVE
ANY QUESTION?**

